NUST CHIP DESIGN CENTRE

C / C++ Programming

Lab # 24
RISC-V SOC Integration with UART

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1. In-Lab Tasks: (*Write your lab task & screenshots here*)

i. Task 1:

Waveform:

2. Critical Analysis: (*Write you critical analysis / conclusion here*)

In this lab I was able to connect my RISC-V processor with a UART transmitter through a memory-mapped interface. At first it was tricky to understand how the special addresses (0x640 for data and 0x644 for status) should behave, but after debugging I confirmed that the store instruction only sends the lower byte when the transmitter is idle, and the load instruction correctly reports the busy flag in the LSB of the destination register. The simulation waveforms showed the UART line going low for the start bit, shifting out the data bits, and then returning high for the stop bit. I also saw the busy signal asserted during transmission and cleared afterward, which matched the expected behavior. This project gave me a much better understanding of how memory-mapped I/O works in a processor, and how peripherals like UART can be controlled using normal load and store instructions.